

# Nomin Erdenebileg

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## EDUCATION

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### University of Illinois Urbana-Champaign

May 2027

Bachelor of Science in Aerospace Engineering

*Relevant Coursework: Mechanics of Composites, UAV Performance, Finite Element Analysis, System Design*

## SKILLS

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Simulation & Analysis    ANSYS Fluent (CFD), ABAQUS (FEA), OpenRocket, Python, MATLAB

Manufacturing            Manual lathes, milling machines, laser cutting, welding, composite layup

CAD & 3D Printing        BambuLab, Onshape, Solidworks, Prusa, Ultimaker

## LEADERSHIP AND ACTIVITIES

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### Independent Project - Flying Wing UAV

August 2026 - Present

- Designing and manufacturing a flying-wing UAV airframe, tackling tailless-configuration aero challenges.
- Ran ABAQUS FEA on wing spars/skins and performance analysis to guide design iterations.

### Aero-Structures Team - SAE Aero Design

March 2026 - Present

- Fabricated a Clark Y airfoil wing; load-tested it and benchmarked results against theoretical values.
- Developed CAD models of structural components for manufacturing and integration.

### Personal Project - High-Power Rocket (Level 1 Certified)

January 2026 - April 2026

- Designed, built, and CAD-modeled L1 carbon-fiber rocket with onboard camera; ran CFD in ANSYS Fluent to optimize aerodynamics.
- Achieved L1 certification, reaching 714 ft apogee (April 2026).

### Airframe Subteam - Liquid Rocketry at Illinois

August 2025 - Present

- Designed and fabricated a roll-control rocket targeting 10,000 ft apogee with two active control canards.
- Load-tested and fabricated fin assemblies; developed supporting CAD models.

### Structures Team - Rocket Propulsion Lab

August 2024 - June 2025

- Fabricated carbon fiber body tubes and nose cones using filament winding and hand layup techniques, and performed tip-to-tip layup on fins.
- Machined aluminum charge wells, bulkheads, and graphite nozzles using manual mills and lathes; also machined fiberglass fins for flight-ready components.
- Performed material analysis to optimize stiffness-to-weight ratio using Hooke's law and selected lightweight structural materials.

## WORK EXPERIENCE

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### Machine Shop Technician - Saint Louis University

August 2024 - May 2025

- Manufactured components using precision manual mills, lathes, and other shop tools to tight tolerances.
- Produced a range of parts including dice, hammers, bulkheads, charge wells, screw switches, fiberglass fins, and solid-propellant casings.
- Operated and maintained shop equipment including band saws, indicators, and other precision tools, emphasizing safety and process consistency.

### Undergraduate Teaching Assistant

Aug 2024 - Present

- Served as TA for Compressible Flow, Statics, and Prototyping courses, supporting instruction, grading, and lab/shop sessions.